

Improving Near-Real-Time Coseismic Landslide Models: Lessons Learned from the 2016 Kaikōura, New Zealand, Earthquake

by Kate E. Allstadt, Randall W. Jibson, Eric M. Thompson, Chris I. Massey,
David J. Wald, Jonathan W. Godt, and Francis K. Rengers

Abstract The U.S. Geological Survey (USGS) is developing near-real-time global earthquake-triggered-landslide products to augment the USGS Prompt Assessment of Global Earthquakes for Response (PAGER) system. The 14 November 2016 M_w 7.8 Kaikōura, New Zealand, earthquake provided a test case for evaluating the performance and near-real-time response applicability of three published global seismically induced landslide models. All three models obtain shaking estimates from the USGS ShakeMap, which is updated and sometimes changes significantly in the hours to days after an earthquake. The Kaikōura earthquake is a particularly valuable event that helps us better understand how changes to the ShakeMap affect the landslide models because the ShakeMap evolved significantly over several weeks as multifault rupture and seismic data were incorporated. We used the detailed landslide inventory available for this event for qualitative landslide model evaluation. We found that once a point source was replaced with an approximate rupture extent in ShakeMap, the landslide models were all successful at roughly identifying the area of highest hazard. This is notable, given that the models are relatively simple, coarse in resolution, and are based solely on input proxies that are globally available. However, all of the models dramatically overpredicted the hazard level, which indicates that improvements can be made. Subsequent updates to the ShakeMap resulted in improvements to model performance by some metrics and declining performance by others. In all cases, details of the ShakeMap strongly controlled the spatial pattern, even when those details were erroneous, such as the inclusion of a fault segment that did not rupture. If maps of landslide hazard are to be used effectively for rapid response, then we need to understand and clearly communicate the control that ShakeMap has over the models and how that typically evolves with time and is (or is not) reflected in reported uncertainties.

Introduction

Landslides triggered by earthquakes pose significant risk to people and infrastructure and create continuing hazards following an earthquake (e.g., [Marano *et al.*, 2010](#)). Rapidly assessing hazards from coseismic landslides is, therefore, an important element in seismic hazard mitigation and the coordination of postseismic response efforts. Most moderate and large earthquakes in mountainous areas trigger landslides. Such landslides commonly account for a significant proportion of coseismic damage and casualties ([Bird and Bommer, 2004](#)), and in some cases, they are the primary source ([Keefer, 1984](#); [Schuster and Fleming, 1986](#)). In addition to direct impacts, landslides block roads and railways, destroy infrastructure, and dam rivers, creating flooding hazards.

To address hazards due to earthquake-induced ground failure, the U.S. Geological Survey (USGS) and collaborators developed a suite of coseismic landslide and liquefaction

models that provide near-real-time estimates of coseismic ground failure hazard based on USGS ShakeMap ground-shaking estimates and other susceptibility proxies ([Allstadt *et al.*, 2016](#)). A rapid estimate of the distribution and extent of coseismic landslide and liquefaction hazard will provide a powerful companion tool to the existing USGS near-real-time products (e.g., ShakeMap and Prompt Assessment of Global Earthquakes for Response [PAGER]) for evaluating damage patterns and response strategies following significant earthquakes. The global scope of existing USGS products requires that the ground failure models be rapidly applicable to any significant earthquake worldwide.

In this study, we focus on the landslide models. The 14 November 2016 M_w 7.8 Kaikōura, New Zealand, earthquake provided a good test case to explore how well the currently implemented global landslide models predicted the actual

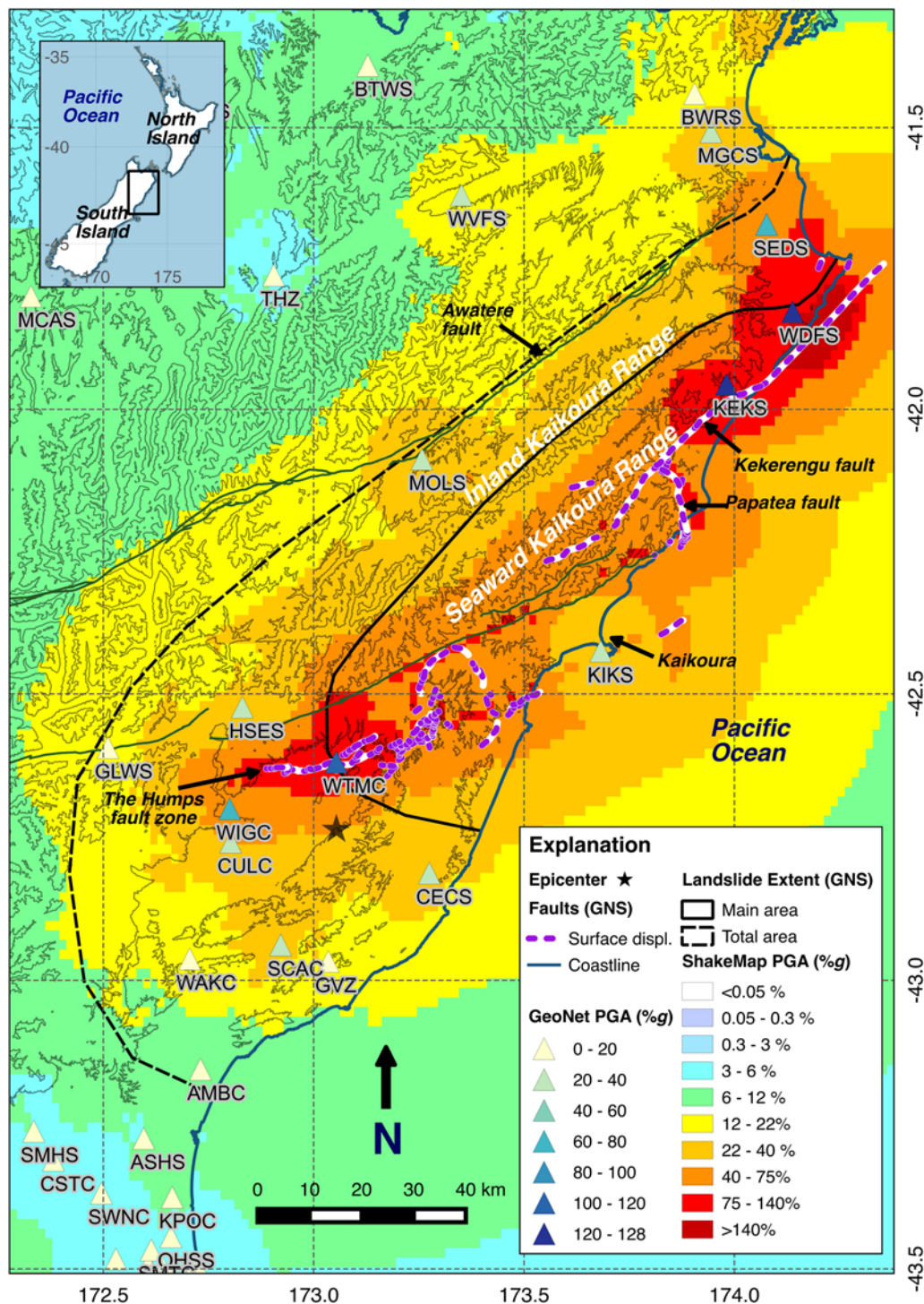


Figure 1. Summary of earthquake ground motions, faulting, and landslide extent. Faults shown (thick dashed line) are those for which surface displacements were measured (Litchfield *et al.*, 2017). Approximate landslide extent, outlined in black, from Massey *et al.* (2018). Peak ground acceleration (PGA) from U.S. Geological Survey (USGS) ShakeMap v. 16 (see Data and Resources). Contours sourced from the Land Information New Zealand (LINZ) data service (see Data and Resources); contours shown start at 300 m with 600 m interval. The color version of this figure is available only in the electronic edition.

landslide hazard and how model performance changed with time after the earthquake. The earthquake struck the northeastern part of the South Island of New Zealand (Fig. 1) and generated surface rupture on more than 20 mapped faults (Litchfield *et al.*, 2018). The earthquake caused widespread

crustal deformation (both uplift and subsidence) and triggered more than 10,000 landslides in the complex topography of an area of about 10,000 km² (Dellow *et al.*, 2017; Jibson *et al.*, 2018; Massey *et al.*, 2018). Landslides triggered by this earthquake formed hundreds of landslide dams, devastated the two

primary transportation routes on the northern part of the South Island (State Highway 1 and the South Island Main Trunk rail line), and at least partially blocked many secondary roads in the region (Jibson *et al.*, 2018).

The USGS mounted a postearthquake field response to assist New Zealand's GNS Science in assessing landslide hazards and to conduct a field evaluation of the outputs of our suite of near-real-time landslide hazard models (Jibson *et al.*, 2018). In this article, we briefly explore these landslide models and evaluate how they performed for the 2016 Kaikōura earthquake. In particular, we examine how the evolution of the ShakeMap ground-motion estimates affected the landslide model predictions. We also suggest needed improvements in both the landslide models and the input parameters on which they depend.

Data and Methods

Field Investigation

Prior to our field visit, we obtained high-resolution satellite imagery covering most of the affected area from DigitalGlobe EV WebHosting (see [Data and Resources](#)) for use in the field. We conducted ground and aerial reconnaissance, both to assess hazards in conjunction with GNS Science and to ground-truth the output of the landslide models. Global Positioning System tracking (~5 m resolution) was used in both ground and aerial reconnaissance to record landslide observations and to navigate to landslide regions identified from postearthquake satellite imagery. Ground surveys were conducted using a four-wheel drive vehicle to access landslide areas, and field observations were documented using handheld cameras. The ground surveys were limited to the edges of the affected area because the existing road access to the interior of the area affected was blocked by landslides. Thus, our ground observations were strongly biased by accessibility. Aerial reconnaissance was done by helicopter, utilizing both handheld cameras and a video camera mounted to the helicopter to document landslide observations (Jibson *et al.*, 2018). Aerial reconnaissance covered all of the primary drainages in the Seaward Kaikōura Range as well as the gentler topography between these mountains and the coast.

Jibson *et al.* (2018) describe field observations and interpretations in detail, including numerous photographs of triggered landslides and other geologic effects of the earthquake. Field observations reported in this article are limited to those relevant to model validation and comparison.

Inventories

Several organizations began compiling landslide inventories using available satellite imagery almost immediately after the earthquake occurred. We used these inventories for guidance in our field investigation, primarily the preliminary version of a polygon inventory compiled by Rathje *et al.* (2017) and a point inventory compiled by Valkaniotis *et al.* (2016).

In the months following the earthquake, GNS Science has compiled a far more detailed inventory of triggered landslides (Massey *et al.*, 2018). Although compilation is ongoing, the current (October 2017) inventory contains more than 10,000 landslides in an affected area of about 10,000 km². Most of the landslides are concentrated in an area of about 3600 km². Mapped landslides range in size (area) from 5 to 550,000 m². Mapping is considered relatively complete for landslide sizes greater than 500 m². Mapping of smaller landslides is incomplete but ongoing (Massey *et al.*, 2018). We use the inventory described by Massey *et al.* (2018) to evaluate our model results.

Landslide Models

Various site-specific and regional approaches to modeling coseismic landslide hazard have been developed (e.g., Carrara *et al.*, 1991; Borga *et al.*, 1998; Guzzetti *et al.*, 1999; Jibson *et al.*, 2000; Guzzetti *et al.*, 2005), but most require detailed inputs or other local data that are inconsistently available on a global scale. In this study, we are restricted to modeling approaches that can be applied rapidly anywhere in the world and thus only use datasets that are globally available in real time. Only a handful of models meet these criteria; they fall into two primary classes. The first class includes physically based models, most of which use a Newmark (1965) sliding-block approach to estimate slope displacement based on a combination of input ground accelerations and temporally stable slope properties, such as topographic slope, material strength, and soil moisture. Resulting displacements are either compared against a threshold to determine whether a slope fails or not, or, alternatively, displacements are related to probability of landslide occurrence (Jibson *et al.*, 2000). The second class of models is statistical. Many use logistic regression to relate landslide occurrence from past earthquakes (inventories) to a group of predictor variables, such as ground shaking, slope, geology, material strength, and soil moisture. Depending on how the inventory data were sampled for the regression, model outputs are depicted as (1) the probability of any landslides occurring in a given cell, (2) the relative hazard for which higher values mean higher hazard without an explicit physical meaning, or (3) the percentage of each cell expected to be affected by landsliding (areal coverage). These models are regressed using landslide inventories, and the meaning of the model output also changes based on whether points or polygon inventories were used and whether only source areas were used or if the entire landslide outline was included.

We are currently able to implement three models on a global scale, the physically based model of Godt *et al.* (2008) and the logistic regression models of Nowicki *et al.* (2014) and Jessee (2017), described in the next paragraph. A few other published models could be applied globally, but we either lack the global input layers required or they do not provide the outputs we require. Kritikos *et al.* (2015) developed a statistical model that theoretically could be applied

Table 1
Summary of Inputs and Methods Used for Each Model

Model	Method	Resolution	Inputs	Outputs
Godt <i>et al.</i> (2008)	Heuristic/Newmark method hybrid	~1 km	ShakeMap PGA*, cohesion [†] , friction angle [†] , and slope quantiles [‡]	Areal coverage [§]
Nowicki <i>et al.</i> (2014)	Logistic regression	~1 km	ShakeMap PGA*, friction angle [†] , maximum slope [‡] , and CTI	Probability of a 1 km pixel experiencing a landslide
Jessee (2017)	Logistic regression	~250 m	ShakeMap PGV*, GMTED calculated slope [#] , GLiM global lithology data**, CTI , and GlobCover 2009 land cover ^{††}	Relative hazard or areal coverage ^{‡‡}

*PGA and peak ground velocity (PGV) sourced from Worden and Wald (2016).

[†]Godt *et al.* (2008).

[‡]Verdin *et al.* (2007).

[§]Percentage of cell affected by landsliding.

^{||}Compound topographic index (CTI) sourced from Moore *et al.* (1991).

[#]Danielson and Gesch (2011).

**Hartmann and Moosdorf (2012).

^{††}Arino *et al.* (2012).

^{‡‡}Relative hazard converted to areal coverage using a transfer function.

globally, but it requires distance to mapped faults. Faults are mapped inconsistently worldwide, and to our knowledge, no global fault map exists. Marc *et al.* (2016) also developed a coseismic landslide model that is globally applicable, but it only provides the total area and volume of landslides without any estimate of how that hazard is spatially distributed.

The first model that we currently implemented is by Godt *et al.* (2008). The model assesses the likelihood of earthquake-induced landslides by combining global topographic data and geologic mapping with near-real-time estimates of ground shaking from ShakeMap. The shear strengths of geological map units are characterized using a global assessment of landslide susceptibility proposed by Nadim *et al.* (2006), which assigns material strength based on age and lithology. Strength parameters and topographic slope are used in a 1D limit-equilibrium slope-stability analysis to calculate the critical acceleration necessary to initiate landsliding. The critical acceleration is then combined with an estimate of strong ground motion from ShakeMap in a simplified Newmark (1965) landslide displacement analysis developed by Jibson (2007), which uses ground-motion parameters instead of time histories to estimate coseismic landslide displacement. The model uses a simplified version of the Newmark (1965) sliding-block method by (1) assuming zero pore pressure, (2) using coarse global geology with representative values of cohesion and friction angle assigned to each unit, (3) using coarse slope quantiles (~1 km resolution based on a 90 m digital elevation model [DEM]), and (4) assuming a semiarbitrary Newmark displacement threshold of 5 cm. The resulting maps show predicted areal coverage of landslides triggered by the input shaking by determining which of the slope quantiles exceeds this threshold in each cell. This method is a heuristic adaptation of the original physically based Newmark method that estimates the proportion of each ~1 km cell that is affected by landsliding (areal coverage). Godt *et al.* (2008) applied the model to three historical earthquakes for which

detailed landslide inventories are available (1976 Guatemala, 1994 Northridge, California, and 2008 Wenchuan, China) and concluded that the model was qualitatively successful at reproducing the patterns of actual landsliding, though no quantitative testing was performed.

Nowicki *et al.* (2014) developed an empirical landslide probability model that uses standardized outputs of ground shaking from the USGS ShakeMap Atlas 2.0 and relatively complete landslide inventories (as defined by Harp *et al.*, 2011) combined with globally available landslide-susceptibility parameters, including topographic slope, surface geology, material strength, and soil wetness. Input parameters are combined in a logistic regression model for which the performance is assessed using statistical goodness-of-fit metrics and a qualitative review to determine which combination of proxies provides both the optimum prediction of landslide-affected areas and minimizes false alarms in nonlandslide zones. The model was calibrated using landslide inventories from four well-documented earthquakes (1976 Guatemala; 1994 Northridge, California; 1999 Chi-Chi, Taiwan; and 2008 Wenchuan, China). Model output is depicted in map form as the probability that a given pixel (~1 km resolution) will be affected by at least one landslide (source area or runoff).

Jessee (2017) expanded on the Nowicki *et al.* (2014) model using a similar approach but including more inventories (23 total). This was done using techniques that allow for the inclusion of incomplete and point inventories, but this complicates the meaning of the resulting probabilities. Jessee (2017) also included new globally available predictor variables (e.g., land cover and improved geology; see Table 1 for details). Another significant difference from the Nowicki *et al.* (2014) model is the use of peak ground velocity (PGV), which was found to be a better predictor variable than peak ground acceleration (PGA), as was used in the Nowicki *et al.* (2014) model. Additionally, the model includes a vegetation-cover layer, and the outputs are higher in

resolution (~250 m compared with ~1 km). Because of the inclusion of incomplete datasets, the model yields relative hazard predictions. However, [Jessee \(2017\)](#) also developed a transfer function to convert the native output to areal coverage. We use the transfer function to convert to areal coverage in this study because it relates to measurable parameters and thus allows us to directly compare the results of the model with those of the mapped inventory.

For all three models, the inventories used for model development and/or testing did not differentiate source areas from runout because they are not commonly mapped separately. This means that the model outputs are representing not just the probability of landslides initiating in a cell but whether or not a cell is likely to be affected by any part of a landslide. Source areas and runout areas have different characteristics (e.g., runout areas do not have to be on slopes), so combining them may also have the effect of diluting the model accuracy, though this effect is likely minimized by the relatively large grid size of the models.

Because all three models were designed for global application, they all are designed to use specific globally available input layers. Given that we are simply comparing previously published models, we do not attempt to modify any of the input layers to obtain better model performance. In this study, we focus on the effects of changing ground-motion estimates. [Allstadt et al. \(2017\)](#) discuss in detail the effect of input layer uncertainties and resolutions.

Ground Motions

As noted in Table 1, all three landslide models use peak ground-motion estimates from the USGS ShakeMap ([Worden and Wald, 2016](#)). ShakeMap is a near-real-time product that provides spatial estimates of PGA, PGV, peak spectral accelerations, and shaking intensities within minutes of any significant earthquake (widely felt and/or $M_w > 5.5$) worldwide. For larger and/or more significant earthquakes, ShakeMap is typically updated numerous times after an event as the magnitude and location are refined. Potential modifications to the ShakeMap include (1) choosing a ground-motion prediction equation (GMPE) to better reflect the mechanism and regional location, (2) incorporating strong ground motion data and citizen-submitted felt reports (“Did You Feel It?”) as they become available and shifting the magnitude used for the GMPE up or down independently for each parameter to better match the observed ground motions (bias correction, [Worden et al., 2010](#)), and (3) adding or updating the estimated rupture geometry (also referred to as a finite fault). Initially, all events are treated as point sources located at the hypocenter, and rupture distances are estimated from epicentral distance and magnitude ([Thompson and Worden, 2017](#)). For larger events with extended ruptures, this can vastly underestimate the extent of strong shaking, so, when available, an estimate of rupture geometry is added to the ShakeMap. In general, the first rupture is based on global seismic inversion ([Hayes et al., 2009](#)) or aftershock extent, but

these can be replaced with more detailed fault representations based on satellite imagery or ground observations if they become available. In practice, updates to the rupture geometry may not occur until long after the event, once research papers are published on the earthquake. Consequently, ShakeMaps are in some cases further refined in retrospect months to years later to include additional information and are included in the ShakeMap Atlas 2.0 ([Garcia et al., 2012](#)) primarily for research purposes. These refined Atlas 2.0 ShakeMaps were used for landslide model development.

GeoNet, the geologic hazard monitoring network for New Zealand, operates four strong ground-motion stations within the main landslide area (see [Data and Resources](#), locations shown in Fig. 1). Three of those had horizontal PGA values that exceeded 1g: (1) station WTMC, which is near the epicenter almost directly on a mapped fault in the south of the Humps fault zone; (2) station KEKS, which is within 3 km of the surface expression of the Kekerengu fault; and (3) station WDFS, which is in the town of Ward about 6 km from the Kekerengu fault. KEKS is on rock; WTMC and WDFS are on deep soil ([Kaiser et al., 2017](#)). The fourth station within the main landslide area is near the town of Kaikōura (KIKS), on rock, and recorded more moderate PGA values of about 0.25g. Vertical accelerations exceeded horizontal PGA at WTMC (~3g) and were about equal to the horizontal accelerations at WDFS. [Bradley et al. \(2017\)](#) noted that durations at the northern stations were much longer than those at stations near the epicenter, which represents time delays between the rupturing of fault segments northward. Duration is considered an important contributor to landslide triggering (e.g., [Newmark, 1965](#)) but is not currently computed by ShakeMap or used in any of the three models. Shaking was likely directed northeastward, as illustrated by broadband ground-motion simulations ([Bradley et al., 2017](#)). This resulted in stronger PGVs in areas lined up along the direction of fault rupture. Directivity is not currently taken into account in ShakeMap, except as reflected in seismic and macroseismic data, and PGA is less affected by directivity than PGV. Therefore, even if directivity is an important control on the triggering of landslides, the only model that would be sensitive to it if it were incorporated in ShakeMap is the [Jessee \(2017\)](#) model.

The near-real-time ShakeMap process for the Kaikōura earthquake was unusual because it took weeks to determine which faults had actually slipped in the earthquake, meaning that refinements to the rupture extent took much longer than usual. The finite-rupture model went through several iterations that began with a single plane derived from teleseismic data ([Hayes et al., 2009](#)) and ended with a detailed version based on geodetic and field observations ([Bradley et al., 2017](#), modified from [Hamling et al., 2017](#)). The GMPE used in the ShakeMap also changed substantially during the process, due to the complexity of the source and uncertainty about whether this was a subduction interface event, a crustal event, or both. There was also some uncertainty about the accuracy of peak ground motions that were measured at stations in the near-fault region, so near-source strong-motion

Table 2
Description of Major ShakeMap Updates for the Kaikōura Earthquake

ShakeMap Version	Time to Release	Description
1	19 min	Automated ShakeMap, M_w 7.4 point source at 10 km depth using Chiou and Youngs (2008) GMPE.
2	1 hr	M_w raised to final value of 7.8, 23 km depth; some distant macroseismic ("Did You Feel It?") data included but no strong-motion data. Changed to Zhao et al. (2006) GMPE for crustal events.
4	3 hrs	Simple single segment rupture derived from global seismic inversion added. More macroseismic data added and some distant strong-motion data, but none in the epicentral region. Changed to Zhao et al. (2006) GMPE for interface events.
6	9 hrs	Strong-motion data from GeoNet seismic stations added, but the four stations nearest to the rupture zone (including the three with $PGA > 1g$) were not yet included.
7	5 days	First version of rupture representation (10 segments) from an early draft of Hamling et al. (2017) and the addition of data from the four stations nearest to the rupture zone. Changed to Bradley (2013) GMPE, specific to New Zealand crustal earthquakes.
16	3 months	Final rupture segmentation model (11 segments) from Bradley et al. (2017) which has several significant changes relative to the previous version, notably the removal of the Awatere fault.

GMPE, ground-motion prediction equation.

data were not included as quickly as usual because it took time to verify those peak values. Table 2 describes the major changes to the ShakeMap versions over time. Versions not listed were primarily minor insignificant tweaks to previous ones. Versions 1–6 reflect the quality of ShakeMaps that are generally released within the first hours after an event when responders typically do not yet know what actually occurred. Version 1 is an automated ShakeMap, and in this case the initial magnitude (M_w 7.4) was lower than the final magnitude (M_w 7.8).

During our field investigations, version 7 was the most up-to-date version and thus was used to guide field reconnaissance. By this time, GNS Science and other responders had already carried out aerial surveys of most of the affected area and knew roughly what had actually occurred and what areas were worst affected. Therefore, our use of the landslide model outputs in the field was primarily an academic exercise. The rupture model used for ShakeMap version 7 was a preliminary version of [Hamling et al. \(2017\)](#) and differed substantially from the most up-to-date rupture model (version 16), which was taken from [Bradley et al. \(2017\)](#), a modification of the final published [Hamling et al. \(2017\)](#) model. In particular, version 7 included a fault north of the affected area, the Awatere fault, that initially was thought to have ruptured but was later determined not to have slipped.

Evaluating Landslide Model Performance

Directly comparing the model outputs, which are in the form of probabilities and areal coverages on grids of differing resolution (Table 1), with the mapped inventory, which is composed of polygons of actual landslides, requires comput-

ing inventory derivatives that are equivalent to each model output. The [Massey et al. \(2018\)](#) inventory separately maps source areas and debris trails. However, for this study, we combine the sources and debris trails for consistency with the inventories that were used to develop the models. All three models used inventories that combined sources and debris trails. To compute the inventory derivative equivalent to the [Nowicki et al. \(2014\)](#) model, which outputs the probability of any landslide in a cell, we simply rasterize the inventory shapefile on the same grid, such that a grid cell is set to one if any part of any landslide polygon lies within that cell. Cells with no part of a landslide polygon within them are set to zero. To compute the inventory derivatives for the other two models, which estimate the percentage of a grid cell's area that will be involved in landsliding, we rasterized the inventory on a grid that vastly over-

sampled the model output grids (~ 1300 samples/km²) where a grid cell is set to one if the center of the cell lies within a landslide polygon and otherwise is set to zero. This was then downsampled using a block mean to the model output grids. This yields a value approximately equal to the total area of each particular grid cell that was part of a landslide.

For all model outputs and their corresponding inventory derivatives, we compute the aggregate hazard (H_{agg}) as a metric to summarize the total hazard estimated by a given model for a given earthquake with a single number. H_{agg} is computed by

$$H_{agg} = \sum_{i=1}^m \sum_{j=1}^n P_{i,j} A,$$

in which $P_{i,j}$ is the probability of a landslide at pixel i, j ; m is the number of rows; n is the number of columns; and A is the pixel/cell area (constant). The ShakeMap extent can differ from version to version and commonly extends far beyond where landslides were likely triggered. To control for this, we only aggregated pixels for which the ShakeMap PGA is greater than or equal to 0.1g, though results are similar without a threshold.

The meaning of the H_{agg} of the inventory is different, and much larger, for the [Nowicki et al. \(2014\)](#) model than for the other two models because the model outputs have differing meanings (Table 1). The [Godt et al. \(2008\)](#) and [Jessee \(2017\)](#) models estimate the percentage of area that experienced a landslide, so H_{agg} has a physical meaning corresponding to the estimated total area of all the landslides within the bounds of the shaking threshold. The [Nowicki et al. \(2014\)](#) model estimates the probability of at least one

landslide occurring in a given cell, which is scale dependent, so H_{agg} is a relative indicator of hazard without a physical meaning. One of the benefits of modeling the percentage area (i.e., coverage) rather than the likelihood of any landslide occurring is that the result does not depend on the resolution of the map. For example, the [Jessee \(2017\)](#) model resolution is a factor of four higher than the [Godt et al. \(2008\)](#) model, and so the areal coverage maps derived from the inventory (inventory derivative) will be different because coverage is being estimated for different cells. However, the H_{agg} computed for each model's respective inventory derivative is the same and is also equal to the sum of the area of all of the landslide polygons of the original inventory that were within the bounds of the shaking threshold. The area is $\sim 20 \text{ km}^2$ in all cases for this event.

Model performance can be judged by a number of criteria. The simplest would be to achieve an accurate H_{agg} . Assessing this is straightforward because we have a detailed field inventory and can compare modeled versus observed (inventory) H_{agg} values. Another evaluation criterion is whether a model accurately identifies locations of high versus low hazard, irrespective of the absolute value of the predictions. To make this evaluation, we quantify how well correlated the predictions of each model are with the observations using normalized cross correlation ([Lewis, 1995](#)) without any offsets. Lastly, the most difficult performance criterion for the models is how well they estimate the hazard level at each given cell (amplitude and spatial accuracy). This can be quantified using predicted versus observed hazard plots. To construct these plots, we classify the observations (inventory derivatives corresponding to each model) into bins of predicted probabilities for each model. The observation associated with each bin is the mean of the observed hazard in each bin. Because cells with lower probabilities are more numerous than those with the highest probabilities, bins are spaced to ensure there are enough entries in each bin to resolve a stable trend. We also used a $0.1g$ shaking threshold for consistency with H_{agg} computation.

Results

The landslide model outputs based on the most up-to-date ShakeMap version 16 (see Fig. 1) are compared with the inventory derivative that is equivalent to the meaning of each model output in Figure 2. The models correctly identify areas that were affected by landsliding but overpredict the extent of landsliding in those areas. They all show high probabilities where landsliding was densest, including within the Seaward Kaikōura Range, in the lower relief areas affected by large deep landslides northeast of Waiau, and along the coastal cliffs that produced landslides that blocked the coastal highway both north and south of Kaikōura. However, the models greatly overestimate landslide probabilities outside these areas of dense landsliding. In reality, landsliding decreased rapidly with distance from the faults that ruptured, with most occurring within 2.5 km of a fault that ruptured ([Massey et al.,](#)

2018). All three models have elevated probabilities extending up to tens of kilometers from the closest actual landslide occurrence in all directions. The statistical models do not have any shaking thresholds in them, and in both cases, small but nonzero probabilities extend beyond the bounds of the shown map. The [Nowicki et al. \(2014\)](#) and [Godt et al. \(2008\)](#) models both predict high hazard in the hills northwest and northeast of Ward, where very few landslides occurred. All models show elevated hazard west of Waiau, also an area of sparse landsliding. Interestingly, all models estimate high hazard along the entirety of the Seaward Kaikōura Range between Ward and Waiau, but the center of the range between Ward and Kaikōura experienced very few triggered landslides. In short, the model probabilities do not decay quickly enough with distance from the fault rupture and produced some particularly stark mismatches in some locations.

The model results shown in Figure 2 are based on ground motions provided by ShakeMap version 16, the most refined version, released about three months after the event when the rupture configuration was better understood. Because the intent of our application of these models is to provide information on landslide hazard to help guide response to the earthquake in the early hours and days before we know what actually occurred, we need to consider what the models would have told us during the response, compared both what they tell us now and what actually occurred. To do this, we ran all of the models for all versions of ShakeMap that were released for this earthquake, starting with the automated version 1. Figure 3 shows the [Nowicki et al. \(2014\)](#) model output for four of these versions that represent major changes to the ShakeMap, as summarized in Table 2. The estimated shaking changes drastically, particularly from version 2 (point source) to version 6 (finite fault). The strongest shaking migrates from around the epicenter, to along the simple finite fault, to much more complex patterns around the multiple fault segments. Starting in version 6 (the first version to include strong ground motion data from the near-source region) bullseyes of higher or lower ground motions around station locations appear, which reflect the radius of influence that data have before decaying back to the underlying GMPE (in this case, [Bradley, 2013](#)). However, the four stations in the main landslide area (described above) were not included in the ShakeMap until version 7, five days after the earthquake, because GeoNet was verifying that the high observed ground motions were accurate.

The models all follow a similar migration as the shaking, from predicting diffuse hazard concentrated around the epicenter to concentrated hazard along the rupture segments. The bullseyes of stronger shaking around the three seismic stations that exceeded $1g$ are also reflected in the landslide models, resulting in higher landslide hazard estimates in sloping areas around those stations than in other locations along the rupture that did not have nearby stations. This pattern was not present in the inventory. Notably, the Awatere fault (Fig. 1) was initially thought to have ruptured and was included in the finite-rupture model for versions 7–15. This

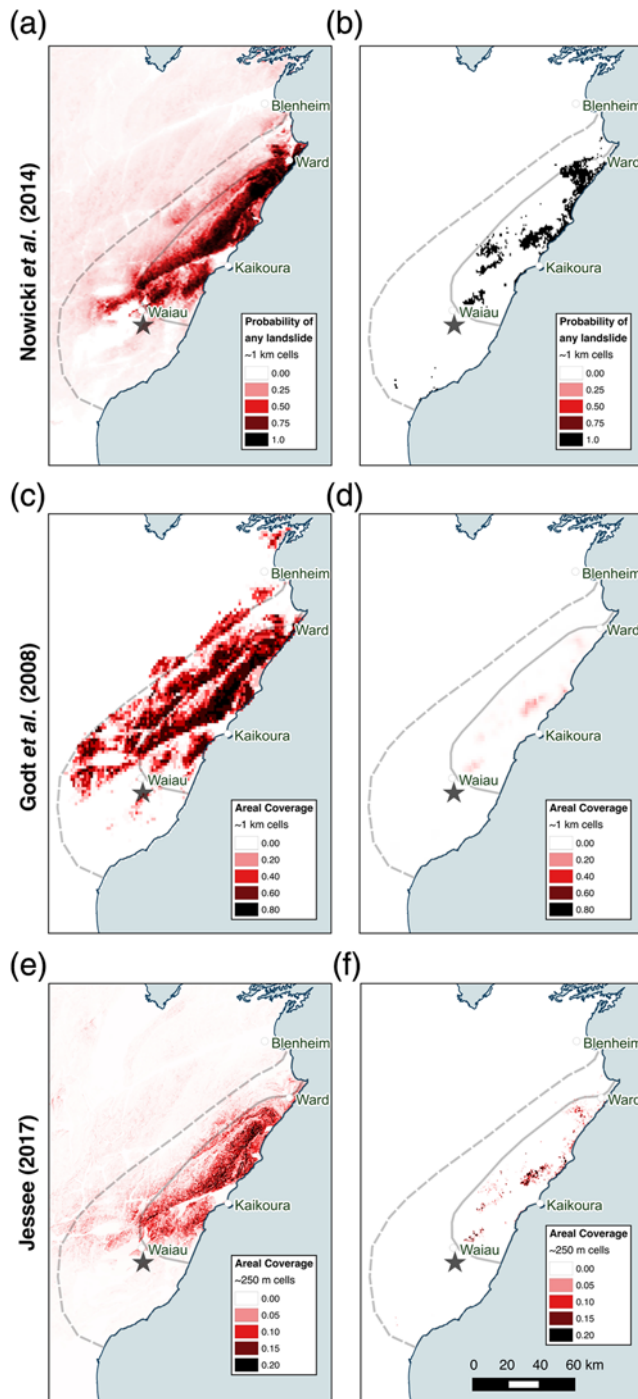


Figure 2. Comparison between (a,c,e) each model and (b,d,f) the Massey *et al.* (2018) inventory converted to simulate the output of each model on the same grid (referred to as an inventory derivative, see the *Evaluating Landslide Model Performance* section for details). The results shown in (c) and (e) are saturated below their peak values (1 and 0.25, respectively) to enhance visibility of the inventory derivative. The color version of this figure is available only in the electronic edition.

resulted in high landslide hazard predicted for slopes along the Awatere Valley to the northeast (Fig. 3c), when in reality almost no landslides occurred there. Version 16 corrected this issue with an updated rupture geometry that did not include the Awatere fault.

Figures 4 and 5 quantify these basic observations by showing the metrics of model performance corresponding to each ShakeMap version as a function of time since the earthquake. Figure 4 shows the evolution of aggregate hazard and the normalized cross correlation with the inventory derivative corresponding to each model described in Table 2. Model uncertainties are shown but account only for ground-motion uncertainties (± 1 standard deviation). None of the landslide models estimate uncertainties related to other aspects of the models, such as uncertainty in their input layers.

All of the models produce H_{agg} results that follow a similar trend and show an overall increase over time, with the exception of a temporary peak when the magnitude was raised to 7.8, and no rupture geometry was incorporated (versions 2 and 3). Interestingly, once the single-plane rupture (Fig. 3b) was incorporated but before the strong-motion data were included (version 4, released about 3 hrs after the earthquake), the H_{agg} of the Nowicki *et al.* (2014) and Jessee (2017) models came close to the actual H_{agg} (Fig. 4b). The GMPE also was changed to the Zhao *et al.* (2006) equation for subduction events at this point because information available at the time suggested the earthquake was on the plate interface. Though H_{agg} was closer to reality for version 4 for all models, the cross-correlation coefficient was still relatively low, particularly for the Jessee (2017) model (Fig. 4b). Simultaneously, the Godt *et al.* (2008) model was more highly correlated with the inventory at this point than it was for the final ShakeMap version (Fig. 4b). The relatively good performance metrics of version 4 should not be interpreted as an indication that the landslide models are more accurate when paired with less accurate estimates of the ground motions; rather, the larger uncertainty in the ground-motion estimates in early versions of the ShakeMap bring down H_{agg} to match the inventory H_{agg} by offsetting errors.

The addition of details obtained from preliminary scientific investigations, including a change from a single-plane rupture to the first version of the multisegment rupture (version 7) and a corresponding change to a New Zealand-specific crustal earthquake GMPE (Bradley, 2013) did not actually improve model performance in any of the metrics for any models. For most models, the cross-correlation coefficient remained about the same as for the single-plane rupture model, and the H_{agg} increased to levels even higher above reality for all models. However, once the Awatere fault was removed and the rupture geometry reconfigured in version 16, all models experienced a steep improvement in inventory correlation to coefficients higher than for the ShakeMap versions with a single rupture plane. This indicates that the larger issue was the rupture geometry rather than the shift to a crustal GMPE.

The estimated uncertainty of all of the models decreases over time (Fig. 4) because the uncertainty of the ShakeMap decreases as rupture geometry is defined and more data are included. In the absence of ground-motion data, the uncertainties come directly from the GMPE. These are then additionally inflated to account for uncertainty in estimating the rupture distances before the rupture is known (Thompson

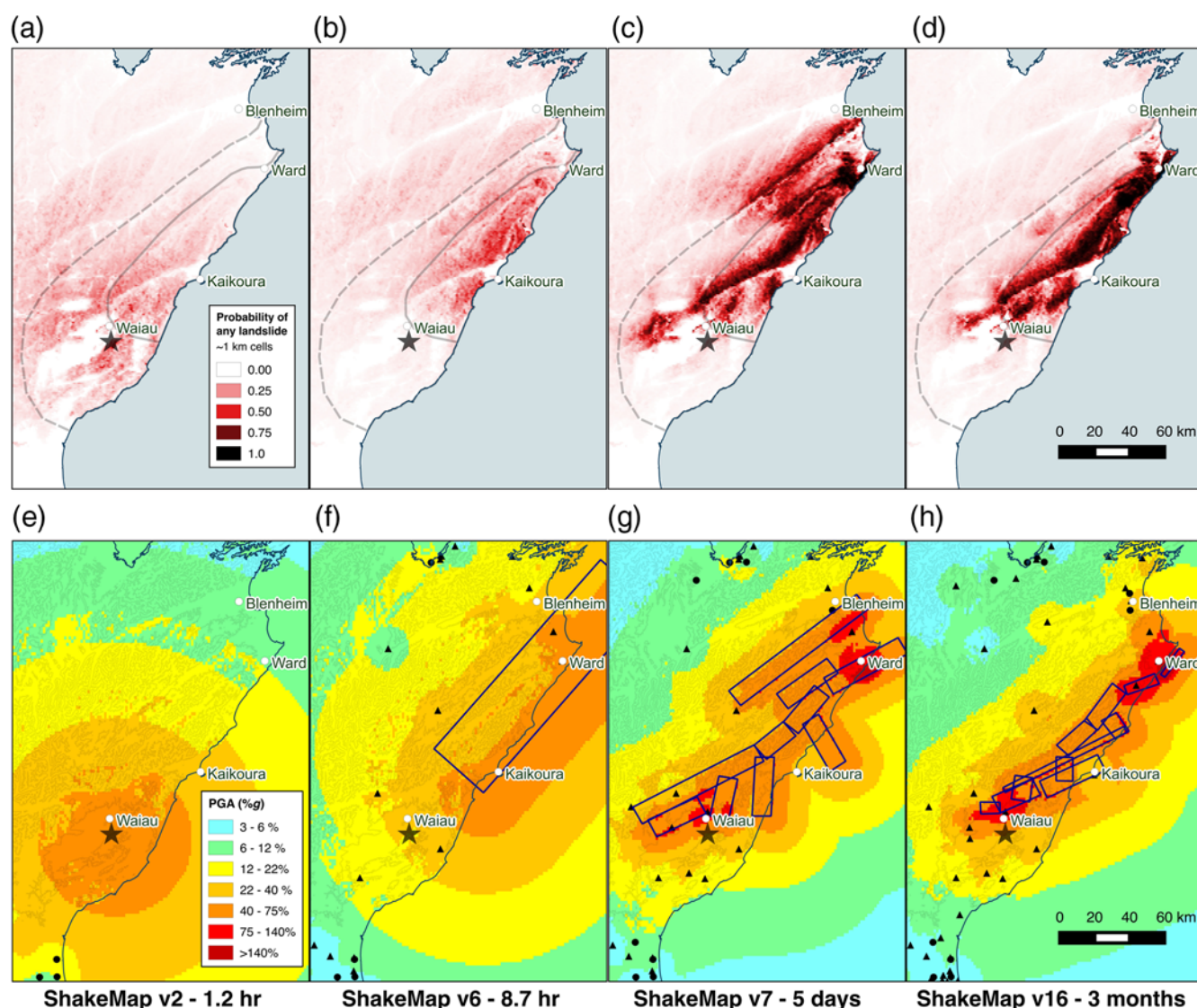


Figure 3. (a–d) Evolution of [Nowicki et al. \(2014\)](#) landslide model with (e–h) four major updates to the USGS ShakeMap, in which the ShakeMap version and release time relative to the earthquake time are labeled below each map; the star indicates the epicenter; triangles and circles show seismic and macroseismic (“Did You Feel It?”) data; and polygons show fault segments used in each version of the ShakeMap. (e) Version 2 uses only a point source at the hypocenter and few data points. (f) Version 6 includes both strong-motion data and a single simple fault based on inversion of teleseismic data. (g) The first estimate of the actual fault configuration based on an early draft of [Hamling et al. \(2017\)](#) that incorporates more data, including strong ground motions in the near-source area. (h) Final adjusted fault configuration based on [Bradley et al. \(2017\)](#). The color version of this figure is available only in the electronic edition.

and Worden, 2017). Before a rupture model or strong-motion data were included (versions 1 and 2), uncertainties on most models spanned at least an order of magnitude. The uncertainties decreased with time after the rupture model was added and later as ground-motion observations were added. In none of the cases is the actual H_{agg} contained within the uncertainty range for the final version once the multifault rupture was added. The [Godt et al. \(2008\)](#) model appears to most drastically overpredict areal coverage, but it actually more closely predicts probability of any landslide in a given cell, which is not its intended output.

H_{agg} and the cross correlation compare the overall hazard level and spatial correlation, respectively, but independ-

ently of each other. The strictest test is whether the models can predict the correct hazard level for each grid cell. This comparison is made in Figure 5, which shows the ratio of a cell-to-cell comparison of the observed hazard (from inventory derivatives) with the predicted hazard. A perfect fit would follow the 1:1 line. In fact, in addition to nearly all of the relationships having a significantly steeper slope than 1:1 (signifying overprediction), there is also a clear offset in each relationship in which the prediction is nonzero and the observed hazard is zero.

The [Nowicki et al. \(2014\)](#) model came the closest to 1:1, though this occurred, not with the final model, but with version 6, which used just the single-plane rupture derived from global seismic inversion and included some strong-mo-

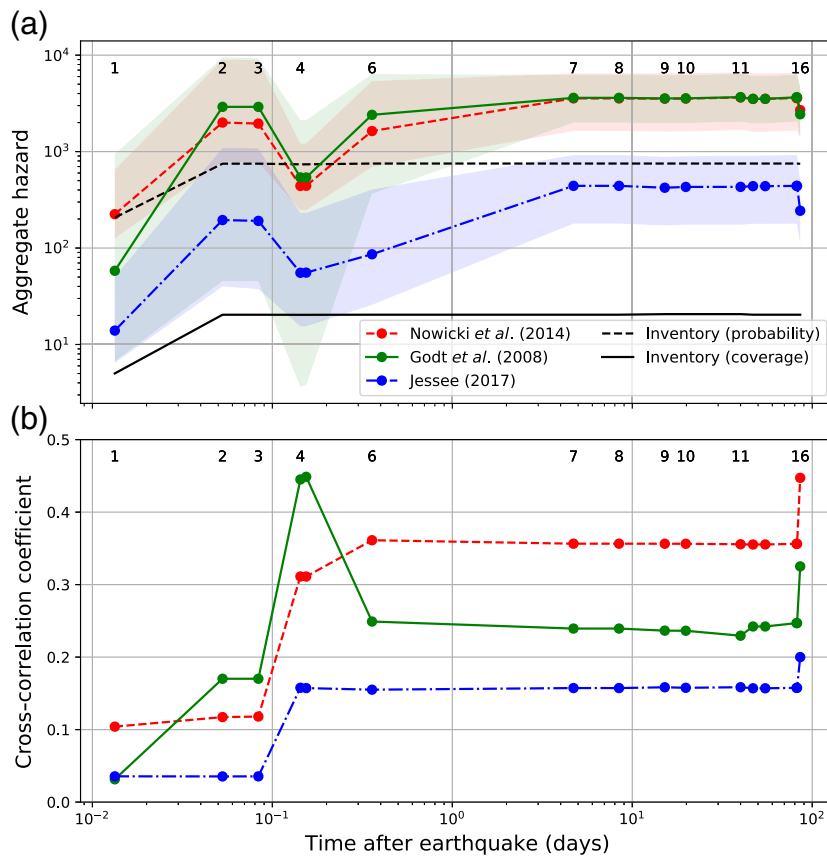


Figure 4. Evolution of model summary statistics with changes in ShakeMap. (a) Changes in aggregate hazard (H_{agg}). Note that the H_{agg} computed from the inventories is stable for both areal coverage and probability, except from version 1 to 2, which is due to a large change in the area that exceeded $0.1g$ in the ShakeMap. (b) Normalized cross-correlation coefficient between each model and its corresponding inventory derivative. The ShakeMap version is labeled at the top of each plot. The color version of this figure is available only in the electronic edition.

tion data, but not in the near-fault region. For this version of ShakeMap, the inventory H_{agg} (Fig. 4a) lies within the range of H_{agg} predicted by the Nowicki et al. (2014) model between ± 1 standard deviation of the ground motions. It seems that adding more detail, at least in this case, decreased the model performance. However, this is not necessarily a general conclusion for all events because the type and quality of data available differ widely from event to event. Adding detail did improve the cross-correlation coefficient by nearly 0.1 units, but it did so at the expense of accurately estimating the hazard level (Fig. 4a).

Discussion

The three models examined all roughly identified the areas of the highest hazard within three hours of the earthquake (ShakeMap version 4). This is despite being relatively simple low-resolution models based on coarse global input layers and no local information, using a ShakeMap with a simple rupture plane derived from teleseismic waveform inversion. Because this is the first relatively well-studied earthquake that triggered significant coseismic landsliding since we implemented these models, we focus in the following sections on lessons learned that can help refine our implementation of near-real-time ground-failure models both in the present and in the long term.

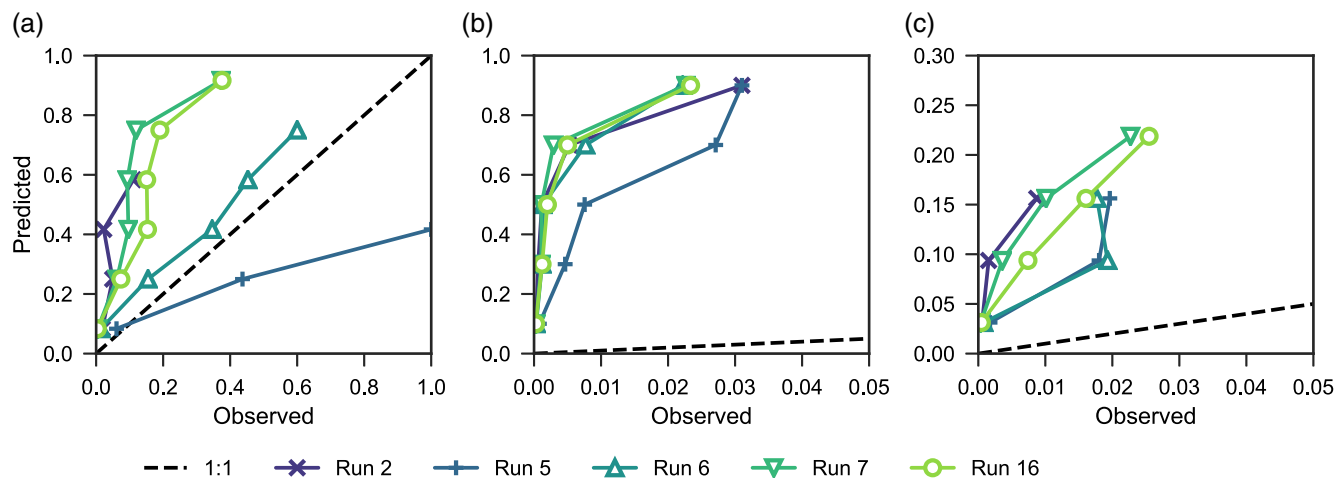


Figure 5. Comparison of the observed versus predicted landslide hazard for (a) Nowicki et al. (2014), (b) Godt et al. (2008), and (c) Jessee (2017). In the Nowicki et al. (2014) model, hazard is represented as the probability of a 1 km pixel experiencing a landslide; the other models portray the areal coverage, which is the percentage of ground failure in each cell. The color version of this figure is available only in the electronic edition.

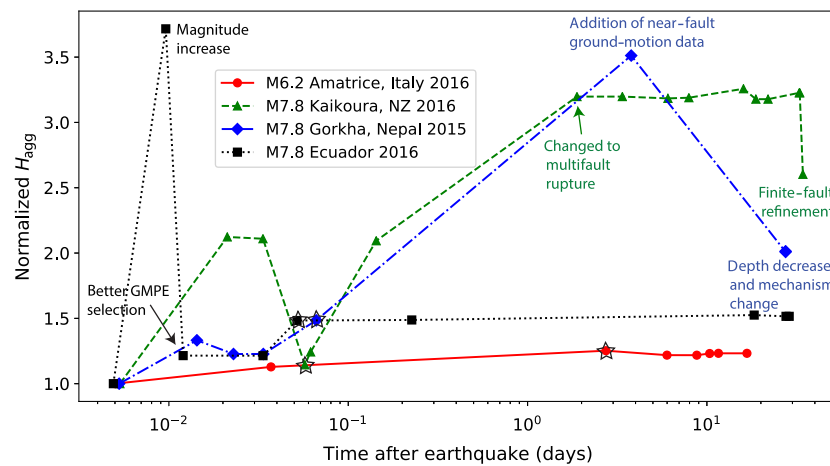


Figure 6. Evolution of landslide aggregate hazard (H_{agg}) estimated by the Nowicki *et al.* (2014) landslide model, normalized by the H_{agg} for the first version, with time after the earthquake. Each subsequent point marks an update to the ShakeMap. The stars indicate the model result corresponding to the first ShakeMap version for which a fault rupture was included for each event. The color version of this figure is available only in the electronic edition.

We can incorporate a few lessons immediately that will improve the usability of these products for near-real-time response efforts, primarily regarding communication. The first is the importance of communicating that the landslide model hazard estimates will evolve as the ShakeMap improves and that users should expect significant changes, especially within the first few hours. This is not an issue that is restricted to only certain events. ShakeMap commonly changes substantially over time as the available input information changes. As a consequence, the H_{agg} of the ground-failure models can change significantly as well. The evolution of H_{agg} is shown for three other earthquakes in Figure 6. The H_{agg} of the three larger earthquakes (all M_w 7.8) fluctuates by more than a factor of 3 relative to the first estimate. The fluctuations are due to a combination of factors (some are indicated in Fig. 6) but primarily because larger earthquakes have larger ruptures. Therefore, they affect larger areas, and even relatively minor changes to the ShakeMap can have a greater impact on H_{agg} . In contrast, the smaller M_w 6.2 Amatrice, Italy, earthquake had a relatively stable H_{agg} (Fig. 6).

As apparent from Figure 3a, a point-source earthquake model results in a landslide model that is poorly correlated with reality for an event such as Kaikōura (Fig. 4b). This suggests that for larger crustal earthquakes that have extended ruptures, we should consider withholding the landslide model results until an extended rupture model is available. Otherwise, depending on how the models are to be used, model outputs based on a point-source earthquake could result in misguided response efforts. Also, though the models were developed using fine-tuned ShakeMaps, such levels of ShakeMap refinement are not generally attained within a time span that would be useful for real-time applications. For the Kaikōura earthquake, the addition of a more accurate rupture geometry model five days later (version 7)

reduced uncertainties estimated by ShakeMap but did not actually improve any of the ground-failure model fits significantly and actually made some worse (Figs. 4 and 5). The best cross-correlation coefficients to date were not achieved until three months later. For this earthquake, government agencies had already carried out aerial surveys of the area and had a good idea of what areas were affected within five days. Thus, the results available within the first day (versions 1–6) would represent what could be used for response in parts of the world and in weather conditions that allow for rapid surveys. In other cases, such as when weather conditions are poor, delayed fine-tuning of the ShakeMap may still be useful for response planning.

We also need to clearly communicate what level of detail and accuracy these models can provide. Most notably, these

are approximate global models that provide a general but imperfect indication of where landsliding is most likely. They do not account for local or regional variations in slope characteristics. One way to communicate model limitations is to report uncertainties. ShakeMap provides uncertainty estimates based on the uncertainties of the GMPE used. These can immediately be incorporated into the landslide models by propagating the uncertainty forward (Fig. 7). When this is done, the spatial pattern remains relatively stable, but the amplitudes change significantly, and therefore H_{agg} can be represented as a probability density function rather than as a single value. This would help communicate the uncertainty behind these global models. If the probability distribution of H_{agg} was computed consistently between earthquakes and models, it could be used rapidly to indicate whether an earthquake is expected to be a big landslide-triggering event and, if so, how bad and how certain. For example, Figure 8 shows the probability that H_{agg} would lie within a given range for each model as it evolved with ShakeMap improvements. This was generated by varying the ground-motion inputs to the landslide models by multiple standard deviations, assuming a normal distribution, and using the cumulative normal distribution to estimate the probability H_{agg} will fall within each bin range. The width of the distribution narrows as fault geometry is constrained and more data are added to the ShakeMap models. For earthquakes without seismic stations in the near-source area, the distribution might never narrow significantly. USGS PAGER issues “traffic-light” alert levels for fatalities and economic losses due to shaking, in which different alert levels correspond to order-of-magnitude bins of fatalities and dollar amounts. PAGER then reports the likelihood of this event falling into each predefined range. We could take a similar approach for ground-failure models by, for example, defining preset H_{agg} alert bins (e.g., green, yellow, red or low,

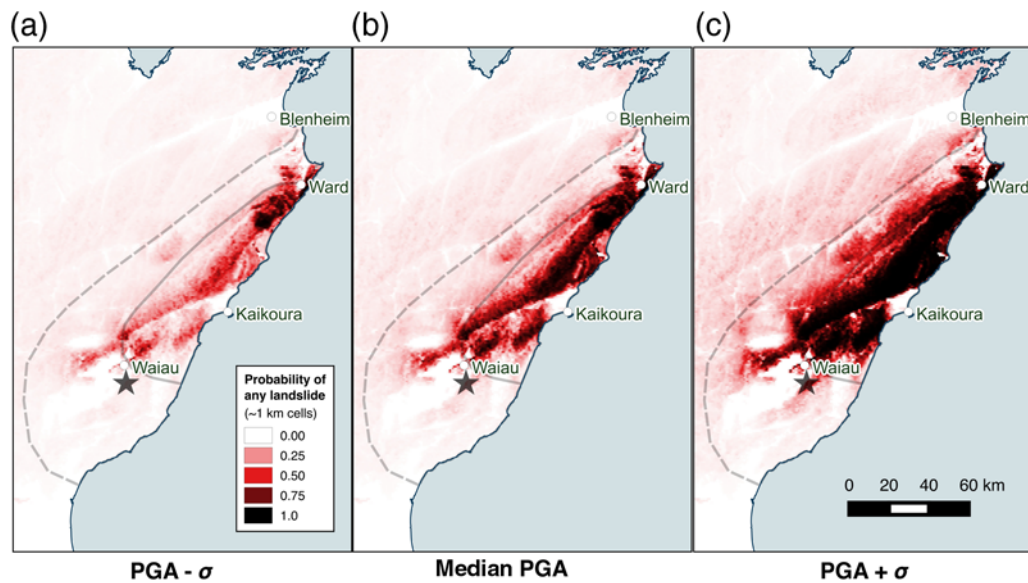


Figure 7. Comparison of Nowicki *et al.* (2014) model output for ShakeMap v. 16 using (a) median PGA minus one standard deviation, (b) median PGA, and (c) median PGA plus one standard deviation. The color version of this figure is available only in the electronic edition.

medium, high) that are calibrated to well-known landslide-triggering events. We could use the uncertainties to show how severe landsliding triggered by a given event is expected to be and how certain we are about that designation.

Current capabilities do not reflect all potential sources of uncertainty. None of the landslide models natively provide any measure of uncertainty. For example, the Godt *et al.* (2008) model depends on slope angle, shear strength (friction angle and cohesion), pore-water pressures, and ground acceleration. However, the model does not account for uncertainties in those input parameters, which can be significant (Dreyfus *et al.*, 2013). Furthermore, the ShakeMap uncertainties do not account for epistemic uncertainties that can arise from errors such as misidentification or mischaracterization of a seismogenic fault. For example, all three models overpredicted hazard substantially along the Inland Kaikōura Range (Fig. 1) when using ground motions from ShakeMap version 7 (Fig. 3c,g) (used during field reconnaissance), because the rupture geometry used at the time included the Awatere fault, a fault that was later determined not to have slipped. This misfit between model prediction and observation demonstrates that, in near-real time, incomplete or inaccurate input data can strongly skew the estimated shaking and ground-failure hazard estimates. Thus, a way of accounting for and communicating this additional uncertainty is needed. The fact that the input was added that was both wrong and strongly affected the shaking estimates (incorrect fault geometry) might also explain why the ShakeMap versions with less-detailed inputs yielded model results in some cases that were nearly as good as or even better by some metrics than the final refined version. Though this was due to circumstances specific to this event, the fact that these earlier models did perform reasonably well when ShakeMap uncertainties are accounted for is encouraging because for many events a ShakeMap similar in detail to version 6, which combines a single-

plane rupture from a global inversion and some shaking data, is as good as we can expect in a time frame short enough to be useful for earthquake response.

Although some of the aforementioned lessons can be acted on without modifying the ground-failure models themselves, some clear lessons from this earthquake should guide future model development. The biggest problem with all the models is that they overpredict hazard. In particular, the model probabilities do not decay quickly enough with distance from the fault-rupture zone. Part of this could be that the models were all trained and/or tested on datasets that included the Wenchuan earthquake. So many landslides were mapped for Wenchuan that even when combined with numerous other inventories, it still contributes the majority of the polygons and can dominate the model (e.g., Tanyas *et al.*, 2017). Therefore, the models may be tuned toward potentially unmodeled details specific to the Wenchuan earthquake (and their respective uncertainties) that might or might not apply to other events in other settings. Yet we still want to be able to identify the next Wenchuan-like event and differentiate it from less drastic but still significant events, such as Kaikōura. This will require a better understanding of the cause of these differences and whether they are related to shaking parameterization, local geology and structure, or other factors that might be accounted for in future model development.

The landslide model overprediction could be due to overestimated ShakeMap ground-motion estimates. However, the evidence suggests that this is not the case. Once strong ground motion records are available, ShakeMap accounts for over- or underestimation by the GMPE by doing a bias correction for each ground-motion parameter. It does this by tuning the earthquake magnitude used in the GMPE up or down to better fit the observations and weighting near-fault data more strongly than distant data (Worden *et al.*, 2010). The bias cor-

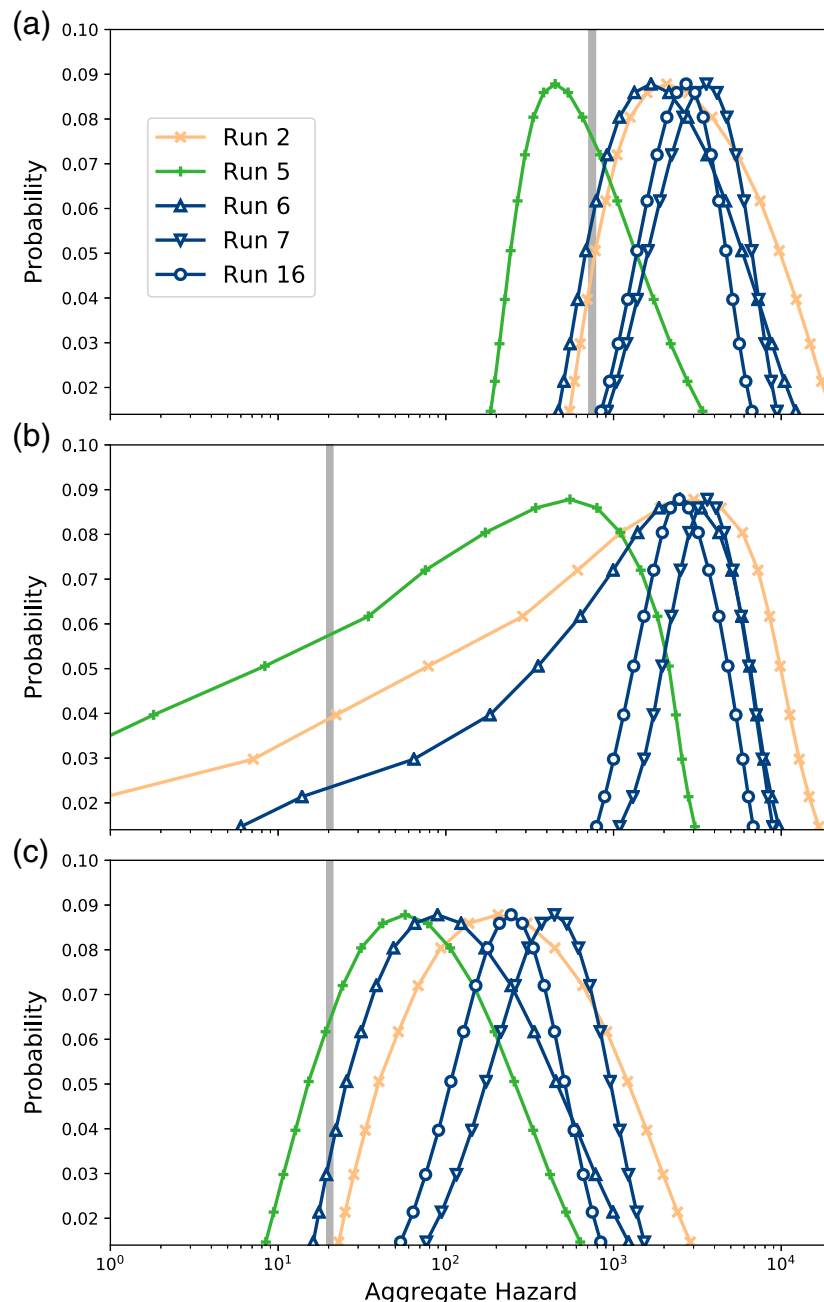


Figure 8. Probability distribution of landslide aggregate hazard (H_{agg}) based on the incorporation of uncertainties in ground-motion input for (a) Nowicki *et al.* (2014), (b) Godt *et al.* (2008), and (c) Jessee (2017) for five different versions of ShakeMap. The ShakeMap for run 2 used point source; run 5 included a finite fault but no strong-motion data; and run 6, 7, and 16 incorporate both a finite-fault model and strong-motion data in the ShakeMap, though version 6 did not yet include any of the near-fault stations with PGAs exceeding 1g. The gray bar indicates the H_{agg} of the inventory derivative corresponding to each model (ground truth). The color version of this figure is available only in the electronic edition.

rection was only -0.03 magnitude units for PGA for Kaikōura (ShakeMap v. 16), which suggests that the shaking was not significantly overestimated, at least at stations within 15 km of a fault that were used in bias correction. The bullseyes of higher shaking around midrange stations such as MOLS, WVFS, MGCS, and SEDS (see Fig. 1) also suggest that

PGA was not overestimated by the GMPE farther from the faults but was actually underestimated. In a practical sense, even if multifault rupture were causing overestimation of shaking, whether or not multifault ruptures similar to this one could be recognized and accurately characterized quickly enough to be useful in near-real-time applications is questionable.

One vexing area of overprediction is the highest peaks of the Seaward Kaikōura Range, where the models predicted high hazard, yet almost no landslides were mapped, despite the steep topography and relatively strong shaking (ground motions exceeded 1g on KEKS, located close to this area). Part of this could be due to the difficulty of mapping new landslides in areas primarily composed of bare rock and scree slopes; however, the largest landslides would have been noticeable, and few were noted. This suggests that there is some change in susceptibility that the model inputs are not currently able to capture. Further investigation using before-and-after DEM differencing might provide insight.

Another pattern of overprediction that stands out is around the seismic stations once their data were incorporated into the ShakeMap. The Nowicki *et al.* (2014) and Godt *et al.* (2008) models both vastly overestimated landslide hazard in the hills due west and northeast of Ward because recorded ground shaking at Ward exceeded 1g, and the ShakeMap algorithm reasonably assumes that if shaking is high at a strong-motion station, it is also high for a certain radius around that station after adjusting for differing site conditions (Worden *et al.*, 2010). Because these two models estimated some of the highest hazard in these areas, we investigated both areas in the field. We expected to find extensive landsliding but found very little. This mismatch is not easily explained, but it is not necessarily the result of an error in the peak ground-motion estimates. Possible explanations include the following: (1) the PGA recorded at Ward could be a highly localized

anomaly, perhaps attributable to site response (though Bradley *et al.*, 2017, did not find pronounced site amplification); (2) although the PGA was high, other ground-motion characteristics such as frequency and duration could be more relevant than peak values and were not conducive to landslide triggering in this area; or (3) geologic conditions in nearby sloping

areas could be mischaracterized in the models (though this does not explain why the model performed better elsewhere).

This artifact suggests that part of the explanation for over-prediction of this event, also invoked by [Massey *et al.* \(2018\)](#), is that PGV and PGA do not sufficiently capture the characteristics of the ground motion that contribute most to landslide triggering. Though high ground accelerations can significantly decrease effective stress on a shear surface and increase downslope driving forces on a potential landslide mass, the frequency and duration of the ground motion also are important ([Jibson *et al.*, 2004](#); [Jibson, 2011](#)). Transients of high-frequency ground acceleration generate little slope displacement because of their short durations ([Newmark, 1965](#)) and thus contribute little to slope-failure probability. Supporting this conclusion, the [Jessee \(2017\)](#) model uses PGV instead of PGA and does not estimate such high probabilities in the area around Ward because PGV is not as sensitive to transient high-amplitude ground accelerations. PGV also has the advantage of not saturating with magnitude as much as PGA does, so it may better reflect the ground motions for larger events. Future iterations of these and other ground-failure models should consider other ground-motion parameters, such as those that account for duration and spectral content because they might be more effective at estimating landslide probabilities. [Jibson \(2007\)](#) suggests Arias intensity ([Arias, 1970](#)), which is the integral of the squared ground acceleration. It implicitly accounts for frequency and duration in addition to amplitude. The challenge is that ShakeMap does not currently output Arias intensity or duration, in part because they are not commonly reported by seismic networks, and so real data are not available for incorporation into ShakeMaps unless USGS has access to the full waveforms. The addition of those parameters and other options for ground-motion parameterization could be explored in future ShakeMap and ground-failure-model development efforts.

During our field reconnaissance, we noted some additional insights that, although likely secondary to the previously discussed issues and not important for every earthquake, are common enough to merit attention in future model development. These observations are detailed in [Jibson *et al.* \(2018\)](#) and are summarized here. One such observation is the coincidence of large landslides and surface ruptures. A total of 44 large landslides occurred on surface ruptures, including the eight largest landslides triggered by the earthquake ([Massey *et al.*, 2018](#)). Many of these large landslides occurred in Tertiary sediment that forms lower-relief gentler topography seaward of the Seaward Kaikōura Range. Surface rupturing appears to have enhanced landslide triggering there ([Massey *et al.*, 2018](#)). This suggests a significant role of ground deformation or some other near-field parameter in the formation of large landslides. Ground deformation is something that is not currently taken into account in the ground-motion or landslide-hazard models. Areas around mapped fault traces could be characterized in the landslide models as having higher landslide susceptibility, but we currently have no way to predict which of such fault traces

would rupture in a given earthquake. Furthermore, faults are mapped at different levels of accuracy and completeness in different parts of the world, so this may result in inconsistencies for global application with current resources.

We also observed that landslides tended to occur preferentially in areas having steep breaks in slope where topographic amplification of ground shaking is likely to have occurred (e.g., [Ashford *et al.*, 1997](#)). For example, steep coastal cliffs and terrace edges seemed to produce higher concentrations of landslides. Material sloughed off from the entire edge of some terraces. Topographic noses at the ends of ridges (convex upward hillslope features) experienced extensive cracking and landsliding. Such concentration of cracking on the ends of ridges has been observed in previous earthquakes ([Keefer and Manson, 1998](#); [Collins and Jibson, 2015](#)). Neither ground-shaking nor landslide susceptibility models currently account for topographic amplification, but the need for shaking models to account for it has long been recognized. Several simplified approaches to approximate topographic amplification have been proposed (e.g., [Maufray *et al.*, 2015](#); [Rai *et al.*, 2017](#)) but have not yet been implemented in ShakeMap.

Conclusion

The Kaikōura earthquake provided an informative test of the ground-failure models we are developing for integration with existing USGS real-time earthquake products (ShakeMap, PAGER, and ShakeCast) for global application. All three coarse global models were able to roughly identify the spatial extent and relative severity of the landsliding triggered by this event, once a rupture plane was incorporated into the ShakeMap (from which the landslide models obtain their shaking estimates). However, all models overestimate the hazard for this event. Their spatial accuracy and the degree to which they overestimate depend heavily on the input ShakeMap ground motions, which evolve and become more refined over time. Because the ShakeMap ground motions evolve to incorporate updated data in the hours to days after the event, so do the ground-failure models. The ground-failure models performed poorly until the earthquake source in ShakeMap was updated from a point source to an extended rupture. This suggests that the ground-failure models should not be released for larger crustal events such as Kaikōura until an extended rupture is incorporated into the ShakeMap. The models performed almost as well using ground motions from the ShakeMap with a simple teleseismically derived rupture plane than a geometry reflecting multifault rupture. Though this may have been a coincidence specific to this event, it is encouraging that the simple model performs reasonably well because simple rupture planes are the best we can expect within the time frame of hours to days for which near-real-time models are most useful for many earthquakes.

The three models tested are purposefully low-resolution coarse models that use input proxies that are consistently available globally. This is because we require the models to be applicable for any significant earthquake worldwide. However, the lack of detailed regional input data means that they

also have significant uncertainties. Communicating these uncertainties clearly is the key to ensuring the model results are used appropriately. This includes quantifying epistemic uncertainties that are currently unaccounted for or underestimated, such as in the model inputs, the models themselves, and the rupture models that are used by ShakeMap.

This study and our field observations also highlighted some additional possibilities for improving future ground-failure models. Better tuning is needed, including identification of factors and globally available proxies that control severity of landsliding that might not currently be included in the existing models so that we can accurately differentiate Wenchuan-like events from Kaikōura-like events from nonevents. We suggest the following improvements to both ShakeMap and ground-failure models: (1) explore the use of additional ground-motion parameters, such as duration and spectral parameters; (2) update the models with higher quality input layers, specifically geological strength parameters; (3) include topographic amplification in shaking estimates; (4) better account for near-fault phenomena and its controls on landsliding; and (5) distinguish source and runout areas in future model development.

Data and Resources

The most recent version of the ShakeMap for the Kaikōura earthquake is available at <https://earthquake.usgs.gov/earthquakes/eventpage/us1000778i#shakemap> (last accessed September 2017). Specific versions of the ShakeMap can be requested using International Federation of Digital Seismograph Networks (FDSN) Event webservices, see <https://earthquake.usgs.gov/fdsnws/event/1/> (last accessed September 2017) for more information. Strong ground motion data for this event can be obtained from GeoNet https://www.geonet.org.nz/data/types/strong_motion (last accessed December 2016). DigitalGlobe satellite imagery is available at <https://evwhs.digitalglobe.com/> (last accessed December 2016). Contours used in map figures are sources from the Land Information New Zealand (LINZ) Data Service, available at <http://data.linz.govt.nz> (last accessed February 2017).

Acknowledgments

This work was funded by the U.S. Geological Survey (USGS) and GNS Science. Simon Cox and GNS Science graciously hosted the USGS field visit and provided extensive logistical support and data sharing. Anna Nowicki Jessee worked with the authors to develop two out of the three models investigated in this study. Kristin Marano and Bruce Worden worked to improve the ShakeMap for this event and to improve ShakeMaps used for model development. Mike Hearne was critical in the development of codes for the implementation of these models in near-real time. The authors also acknowledge the thoughtful reviews from Kristin Marano, Sam McColl, and an anonymous reviewer, as well as feedback from Associate Editor Timothy Stahl, all of which significantly improved the article.

Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

References

- Allstadt, K. E., E. M. Thompson, M. Hearne, M. A. Nowicki Jessee, J. Zhu, D. J. Wald, and H. Tanyas (2017). Integrating landslide and liquefaction hazard and loss estimates with existing USGS real-time earthquake information products, *Proc. of the 16th World Conf. on Earthquake Engineering*, Santiago, Chile, 9–13 January 2017, 13 pp.
- Allstadt, K. E., E. M. Thompson, D. J. Wald, M. W. Hamburger, J. W. Godt, K. L. Knudsen, R. W. Jibson, M. A. Jessee, J. Zhu, M. Hearne, et al. (2016). USGS approach to real-time estimation of earthquake-triggered ground failure—Results of 2015 workshop, *U.S. Geol. Surv. Open-File Rept. 2016-1044*, 13 pp.
- Arias, A. (1970). A measure of earthquake intensity, in *Seismic Design for Nuclear Power Plants*, R. J. Hansen (Editor), Massachusetts Institute of Technology Press, Cambridge, Massachusetts, 438–483.
- Arino, O., R. Perez, J. Julio, V. Kalogirou, S. Bontemps, P. Defourny, and E. Van Bogaert (2012). *Global Land Cover Map for 2009 (GlobCover 2009)*, European Space Agency (ESA) & Université catholique de Louvain (UCL), doi: [10.1594/PANGAEA.787668](https://doi.org/10.1594/PANGAEA.787668).
- Ashford, S. A., N. Sitar, J. Lysmer, and N. Deng (1997). Topographic effects on the seismic response of steep slopes, *Bull. Seismol. Soc. Am.* **87**, 701–709.
- Bird, J. F., and J. J. Bommer (2004). Earthquake losses due to ground failure, *Eng. Geol.* **75**, 147–179.
- Borga, M., G. Dalla Fontana, D. Da Ros, and L. Marchi (1998). Shallow landslide hazard assessment using a physically based model and digital elevation data, *Environ. Geol.* **35**, 81–88.
- Bradley, B. A. (2013). A New Zealand-specific pseudospectral acceleration ground-motion prediction equation for active shallow crustal earthquakes based on foreign models, *Bull. Geol. Soc. Am.* **103**, 1801–1822.
- Bradley, B. A., H. N. Razafindrakoto, and V. Polak (2017). Ground-motion observations from the 14 November 2016 M_w 7.8 Kaikōura, New Zealand, earthquake and insights from broadband simulations, *Seismol. Res. Lett.* **88**, 740–756.
- Carrara, A., M. Cardinali, R. Detti, F. Guzzetti, V. Pasqui, and P. Reichenbach (1991). GIS techniques and statistical models in evaluating landslide hazard, *Earth Surf. Process. Landf.* **16**, 427–445.
- Chiou, B. J., and R. R. Youngs (2008). An NGA model for the average horizontal component of peak ground motion and response spectra, *Earthq. Spectra* **24**, no. 1, 173–215.
- Collins, B. D., and R. W. Jibson (2015). Assessment of existing and potential landslide hazards resulting from the April 25, 2015 Gorkha, Nepal, earthquake sequence, *U.S. Geol. Surv. Open-File Rept. 2015-1142*, 50 p.
- Danielson, J., and D. Gesch (2011). Global multi-resolution terrain elevation data 2010 (GMTED2010), *U.S. Geol. Surv. Open-File Rept. 2011-1073*, 23 pp.
- Dellow, S., C. Massey, S. Cox, G. Archibald, J. Begg, Z. Bruce, J. Carey, J. Davidson, F. Della Pasqua, P. Glassey, et al. (2017). Landslides caused by the M_w 7.8 Kaikōura earthquake and the immediate response, *Bull. New Zeal. Soc. Earthq. Eng.* **50**, 106–116.
- Dreyfus, D., E. M. Rathje, and R. W. Jibson (2013). The influence of different simplified sliding-block models and input parameters on regional predictions of seismic landslides triggered by the Northridge earthquake, *Eng. Geol.* **163**, 41–54.
- Garcia, D., R. T. Mah, K. L. Johnson, M. G. Hearne, K. D. Marano, K. W. Lin, D. J. Wald, C. B. Worden, and E. So (2012). ShakeMap Atlas 2.0: An improved suite of recent historical earthquake ShakeMaps for global hazard analyses and loss models, *Proc. of the 15th World Conf. on Earthquake Engineering*, Lisbon, Portugal, 10 pp.
- Godt, J. W., B. Şener, K. L. Verdin, D. J. Wald, P. S. Earle, E. L. Harp, and R. W. Jibson (2008). Rapid assessment of earthquake-induced landsliding, *Proc. of the 1st World Landslides Forum*, United Nations University, Tokyo, Japan, 219–222.
- Guzzetti, F., A. Carrara, M. Cardinali, and P. Reichenbach (1999). Landslide hazard evaluation: A review of current techniques and their application in a multi-scale study, Central Italy, *Geomorphology* **3**, 181–216.
- Guzzetti, F., P. Reichenbach, M. Cardinali, M. Galli, and F. Ardizzone (2005). Probabilistic landslide hazard assessment at the basin scale, *Geomorphology* **72**, 272–299.

- Hamling, I. J., S. Hreinsdottir, K. Clark, J. Elliott, C. Liang, E. Fielding, N. Litchfield, P. Villamor, L. Wallace, T. J. Wright, *et al.* (2017). Complex multifault rupture during the 2016 M_w 7.8 Kaikōura earthquake, New Zealand, *Science* doi: [10.1126/science.aam7194](https://doi.org/10.1126/science.aam7194).
- Harp, E. L., D. K. Keefer, H. P. Sato, and H. Yagi (2011). Landslide inventories: The essential part of seismic landslide hazard analyses, *Eng. Geol.* **122**, 9–21.
- Hartmann, J., and N. Moosdorf (2012). The new global lithological map database GLiM: A representation of rock properties at the Earth surface, *Geochim. Geophys. Res.* **13**, no. 12, doi: [10.1029/2012GC004370](https://doi.org/10.1029/2012GC004370).
- Hayes, G. P., L. Rivera, and H. Kanamori (2009). Source inversion of the W phase: Real-time implementation and extension to low magnitudes, *Seismol. Res. Lett.* **80**, 817–822, doi: [10.1785/gssrl.80.5.817](https://doi.org/10.1785/gssrl.80.5.817).
- Jessee, M. A. (2017). An investigation of seismic hazard: Using geophysical data to predict the location and impact of earthquake-induced landslides around the globe and to estimate earthquake potential in the Philippines, *Ph.D. Dissertation*, Indiana University.
- Jibson, R. W. (2007). Regression models for estimating coseismic landslide displacement, *Eng. Geol.* **91**, 209–218.
- Jibson, R. W. (2011). Methods for assessing the stability of slopes during earthquakes—A retrospective, *Eng. Geol.* **122**, 43–50.
- Jibson, R. W., K. E. Allstadt, F. K. Rengers, and J. W. Godt (2018). Overview of the geologic effects of the November 14, 2016, M_w 7.8 Kaikōura, New Zealand, earthquake, *U.S. Geol. Surv. Sci. Investig. Rept. 2017-5146*, 39 pp., doi: [10.3133/sir20175146](https://doi.org/10.3133/sir20175146).
- Jibson, R. W., E. L. Harp, and J. A. Michael (2000). A method for producing digital probabilistic seismic landslide hazard maps, *Eng. Geol.* **58**, 271–289.
- Jibson, R. W., E. L. Harp, W. Schulz, and D. K. Keefer (2004). Landslides triggered by the 2002 Denali fault, Alaska, earthquake and the inferred nature of the strong shaking, *Earthq. Spectra* **20**, 669–691.
- Kaiser, A., N. Balfour, B. Fry, C. Holden, N. Litchfield, M. Gerstenberger, E. D'Anastasio, N. Horspool, G. McVerry, J. Ristau, *et al.* (2017). The 2016 Kaikōura, New Zealand, earthquake: Preliminary seismological report, *Seismol. Res. Lett.* **88**, no. 3, 1938–2057, doi: [10.1785/0220170018](https://doi.org/10.1785/0220170018).
- Keefer, D. K. (1984). Landslides caused by earthquakes, *Bull. Geol. Soc. Am.* **95**, 406–421.
- Keefer, D. K., and M. W. Manson (1998). Regional distribution and characteristics of landslides generated by the earthquake, in *The Loma Prieta, California, Earthquake of October 17, 1989—Landslides*, D. K. Keefer (Editor), *U.S. Geol. Surv. Profess. Pap. 1551-C*, C7–C32.
- Kritikos, T., T. R. Robinson, and T. R. H. Davies (2015). Regional coseismic landslide hazard assessment without historical landslide inventories: A new approach, *J. Geophys. Res.* **120**, no. 4, 711–729, doi: [10.1002/2014JF003224](https://doi.org/10.1002/2014JF003224).
- Lewis, J. P. (1995). Fast normalized cross-correlation, *Vis. Interface* **10**, 120–123.
- Litchfield, N. J., A. Benson, A. Bischoff, A. Hatem, A. Barrier, A. Nicol, A. Wandres, B. Lukovic, B. Hall, C. Gasston, *et al.* (2017). 14th November 2016 M7.8 Kaikōura earthquake—Summary surface fault rupture traces and displacement measurements, Item details—188412, *GNS Science*, doi: [10.21420/G2RC7C](https://doi.org/10.21420/G2RC7C).
- Litchfield, N. J., P. Villamor, R. J. Van Dissen, A. Nicol, P. M. Barnes, D. J. A. Barrell, J. Pettinga, R. M. Langridge, T. A. Little, J. Mountjoy, *et al.* (2018). Surface fault rupture from the M_w 7.8 2016 Kaikōura earthquake, New Zealand, and insights into factors controlling multi-fault ruptures, *Bull. Seismol. Soc. Am.* (this issue).
- Marano, K. D., D. J. Wald, and T. I. Allen (2010). Global earthquake casualties due to secondary effects: A quantitative analysis for improving rapid loss analyses, *Nat. Hazards* **52**, 319–328, doi: [10.1007/s11069-009-9372-5](https://doi.org/10.1007/s11069-009-9372-5).
- Marc, O., N. Hovius, P. Meunier, T. Gorum, and T. Uchida (2016). A seismologically consistent expression for the total area and volume of earthquake-triggered landsliding, *J. Geophys. Res.* **121**, no. 5, 640–663, doi: [10.1002/2015JF003732](https://doi.org/10.1002/2015JF003732).
- Massey, C., D. Townsend, E. Rathje, K. Allstadt, Y. Kaneko, B. Lukovic, B. Bradley, J. Wartman, N. Horspool, I. Hamling, *et al.* (2018). Landslides triggered by the 14 November 2016, M_w 7.8 earthquake, Kaikōura, New Zealand, *Bull. Seismol. Soc. Am.* (this issue).
- Maufroy, E., V. M. Cruz-Atienza, F. Cotton, and S. Gaffet (2015). Frequency-scaled curvature as a proxy for topographic site-effect amplification and ground-motion variability, *Bull. Seismol. Soc. Am.* **105**, no. 1, 354–367, doi: [10.1785/0120140089](https://doi.org/10.1785/0120140089).
- Moore, I. D., R. B. Grayson, and A. R. Ladson (1991). Digital terrain modelling: A review of hydrological, geomorphological, and biological applications, *Hydrol. Process.* **5**, no. 1, 3–30.
- Nadim, F., O. Kjekstad, P. Peduzzi, C. Herold, and C. Jaedicke (2006). Global landslide and avalanche hotspots, *Landslides* **3**, 159–173.
- Newmark, N. M. (1965). Effects of earthquakes on dams and embankments, *Geotechnique* **15**, 139–159.
- Nowicki, M. A., D. J. Wald, M. W. Hamburger, M. Hearne, and E. M. Thompson (2014). Development of a globally applicable model for near-real-time prediction of seismically induced landslides, *Eng. Geol.* **173**, 54–65.
- Rai, M., A. Rodriguez-Marek, and B. S. Chiou (2017). Empirical terrain-based topographic modification factors for use in ground motion prediction, *Earthq. Spectra* **33**, no. 1, 157–177.
- Rathje, E. M., M. Little, J. Wartman, A. G. Athanasopoulos-Zekkos, C. I. Massey, and N. Sitar (2017). Preliminary landslide inventory for the 2016 Kaikōura, New Zealand earthquake derived from satellite imagery and aerial/field reconnaissance, version 1, 4 January 2017, Geotechnical Extreme Events Reconnaissance (GEER) Association, doi: [10.18118/G6NK57](https://doi.org/10.18118/G6NK57).
- Schuster, R. L., and R. W. Fleming (1986). Economic losses and fatalities due to landslides, *Bull. Assoc. Eng. Geol.* **23**, 11–28.
- Tanyas, H., C. J. van Westen, K. E. Allstadt, M. A. Nowicki-Jessee, T. Gorum, R. W. Jibson, J. W. Godt, H. P. Sato, R. G. Schmidt, O. Marc, *et al.* (2017). Presentation and analysis of a worldwide database of earthquake-induced landslide inventories, *J. Geophys. Res.* **122**, no. 10, 1991–2015, doi: [10.1002/2017JF004236](https://doi.org/10.1002/2017JF004236).
- Thompson, E. M., and C. B. Worden (2017). Estimating rupture distances without a rupture, *Bull. Seismol. Soc. Am.* doi: [10.1785/0120170174](https://doi.org/10.1785/0120170174).
- Valkaniotis, S., G. Papathanassiou, and S. Pavlides (2016). Preliminary map of coseismic landslides for the M 7.8 Kaikōura, New Zealand earthquake, *Zenodo* doi: [10.5281/zenodo.167130](https://doi.org/10.5281/zenodo.167130).
- Verdin, D. W., J. Godt, C. Funk, D. Pederos, B. Worstell, and J. Verdin (2007). Development of a global slope dataset for estimation of landslide occurrence resulting from earthquakes, *U.S. Geol. Surv. Open-File Rept. 2007-1188*.
- Worden, C. B., and D. J. Wald (2016). ShakeMap manual online: Technical manual, user's guide, and software guide, *U.S. Geol. Surv.* doi: [10.5066/F7D21VPQ](https://doi.org/10.5066/F7D21VPQ).
- Worden, C. B., D. J. Wald, T. I. Allen, K. Lin, D. Garcia, and G. Cua (2010). A revised ground-motion and intensity interpolation scheme for ShakeMap, *Bull. Seismol. Soc. Am.* **100**, 3083–3096.
- Zhao, J. X., J. Zhang, A. Asano, Y. Ohno, T. Ouchi, T. Takahashi, H. Ogawa, K. Irikura, H. K. Thio, P. G. Somerville, *et al.* (2006). Attenuation relations of strong ground motion in Japan using site classification based on predominant period, *Bull. Seismol. Soc. Am.* **96**, no. 3, 898–913.

U.S. Geological Survey
Geologic Hazards Science Center
P.O. Box 25046, DFC, MS 966
Denver, Colorado 80225
kallstadt@usgs.gov
(K.E.A., R.W.J., E.M.T., D.J.W., J.W.G., F.K.R.)

GNS Science
1 Fairway Drive, Avalon 5010
P.O. Box 30-368
Lower Hutt 5040, New Zealand
(C.I.M.)

Manuscript received 5 October 2017;
Published Online 27 March 2018;
Corrected Online 14 May 2018